that are not too small be obtained—a fraction of the peak value of the input—it is also necessary that the discharge time constant (R_dC) be several times the period of the applied pulses.

Acknowledgment

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Low-Frequency Shielding of a Circular Loop Electromagnetic Field Source

J. RONALD MOSER

Abstract—**The problem of low-frequency shielding of a loop axially perpendicular to a plane shield of infinite extent is analyzed by 1) the thin shield work of S. Levy, 2) solution of the vector wave equation, and 3) application of the transmission theory of shielding of Schelkunoff.**

Experimental data are obtained and compared with results of parts 1) and 3) in the frequency range 100 Hz to 50 kHz. The first analytical technique is not general, and the limits of applicability of the results are discussed.

In the second solution, which is general, expressions are derived for the total electric and magnetic fields on both sides of and within the shield. The resulting expression for shielding effectiveness is not solved because of its complexity.

The results of the third theory are adapted to the problem. The shielding effectiveness expression $S = R + A + B$ is computer evaluated for the six shields considered (1/16-inch and 1/8-inch thick aluminum, copper, and steel). Although some approximations are made, this analytical method is the most useful in predicting the insertion loss of the shield, since the theory includes those parameters neglected in the first analytical technique.

Introduction

THERE ARE numerous techniques of analysis used in the solution of shielding problems. Three analytical techniques are considered here in solving a shielding problem of fundamental importance: low-frequency electromagnetic shielding of a circular loop field source by a thin plane metal sheet of infinite extent. The first technique discussed is a theoretical determination of the fields by considering the induced eddy currents.[1] The second method is an exact solution of the vector wave equation.[2],[3] The third method is an extension of the transmission theory of shielding.[4],[5] Each analytical method has advantages and disadvantages for different classes of shielding problems. The first and second methods are easily applied only in the limited class of problems where the geometry of the problem fits one of the natural coordinate systems. Cylindrical coordinates will be applicable to the problem at hand. The third method is of great value to the large class of plane wave shielding problems. An adaptation of this theory to the present problem is included. This theory then provides a definite approximation procedure when the previous analytical methods fail. Experimentally obtained data are used to check the results of the first method. The plane wave transmission theory of shielding is adapted to the problem by comparing the computer evaluated results of this theory with these data.

Another technique (not considered here) consists of expressing the circulating eddy currents within the shield in a multipole expansion. The reader is referred to the work of the Boeing Company.[6]

Throughout this study, unless otherwise noted, shielding effectiveness of a particular shield will be defined as the flux density (or voltage) insertion loss, in decibels, created by the introduction of the shield between a loop-field source and a loop-field sensor.

Theory

The first analytical method to be discussed is based on the classical eddy-current thin shield work of Maxwell. Levy adapted this analysis to this problem in 1933. He formulated the magnetic potentials and satisfied boundary

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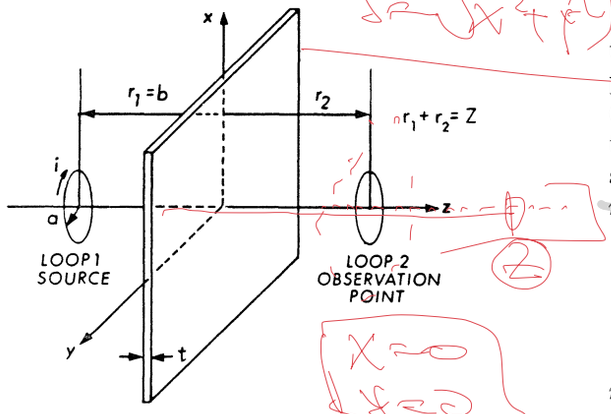


Fig. 1. Coaxial loops separated by an infinite plate.

conditions by postulating moving images. A discussion of his results follows. See Fig. 1.

The total voltage induced in loop 1 due to eddy currents in the shield is

$$V_1' = M(2r_1)I\omega \cos\omega t + \int_{2r_1}^{\infty} \frac{dM(z)}{dz} I\omega \cos\omega \left(t - \frac{z - 2r_1}{V}\right) dz. \quad (1)$$

To expedite integration, Levy then formed an approximate equation[1] for $M(z)$:

$$M(z) = 28.6\, n^2 e^{-9z/5a} 10^{-9} \text{ henrys}. \quad (2)$$

This approximation will limit the solution to ten percent accuracy within the limits $0.15 < r_1/a < 0.9$. Other approximate forms of $M(z)$ could be derived from existing tables to extend the range of applicability of this solution.

Making this substitution in V_1' of (1) and performing the integration, Levy obtained

$$V_1' = -I(R' \sin\omega t + L'\, \omega \cos\omega t) \quad (3)$$

where

$$R' = \frac{1.42\, n^2\, 10^{-5}\, (1/\sigma_r t) e^{-3.6r_1/a}}{1 + (78.8/at\ \sigma_r f)^2} \quad (4)$$

and

$$L' = \frac{-28.6\, n^2 10^{-9}\, e^{-3.6r_1/a}}{1 + (78.8/at\ \sigma_r f)^2} \quad (5)$$

where

R', L'	the apparent changes in real and imaginary parts of the impedance of loop 1
n	number of turns in loop 1
σ_r	percent conductivity of the shield material
t	shield thickness, cm
r_1	distance in cm from loop 1 to shield (see Fig. 1)
a	radius of loop 1, cm
f	frequency, Hz.

[1] See references 6 and 7 in the bibliography of the paper by Levy.[1]

To solve the actual shielding problem, Levy assumed the eddy currents to be distributed such that the induced potential on both sides of the shield would be identical. (Note that this would seem to restrict validity of results to shields of thickness less than a skin depth.) With this assumption, and by replacing $2r_1$ by $r_1 + r_2$ and n^2 by nn' in (3), one obtains, for the voltage induced in loop 2,

$$V_2' = \frac{-1.42(1/\sigma_r t)\ 10^4 - j28.6\omega}{1 + (78.8/at\sigma_r f)^2}\, nn' 10^{-9} e^{-1.8(r_1+r_2)/a}\, I.$$

It is important to note that V_2' is independent of r_1 or r_2 taken separately; that is, V_2' will be independent of the position of the shield between the two loops as long as the sum $r_1 + r_2$ is unchanged.

With the shield removed, the voltage in loop 2 is

$$V_2'' = -j\omega M_{12} I = -j\omega 28.6 ann' 10^{-9} I e^{-9(r_1+r_2)/5a}.$$

Adding the two opposing phasor voltages V_2' and V_2'' to obtain the total voltage induced in loop 2 yields

$$V_2 = V_2' + V_2''. \quad (6)$$

Then the insertion loss of the shield is

$$\left|\frac{V_2''}{V_2}\right| = \left|\frac{28.6\omega a[1 + (78.8/at\sigma_r f)^2]}{-1.42(1/\sigma_r t)10^4 - j(78.8/at\sigma_r f)^2 \omega 28.6a}\right|$$

$$= [(at\sigma_r f/78.8)^2 + 1]^{1/2}.$$

The voltage drop (in decibels) caused by introducing the shield between the two loops is therefore

$$S \equiv 20 \log_{10}\left|\frac{V_2''}{V_2}\right| = 10 \log_{10}[(at\sigma_r f/78.8)^2 + 1] \quad (7)$$

where the symbols have the same meaning as in (4) and (5), and the accuracy is still dependent on the approximations made in forming $M(z)$.

Note that S as given here is apparently independent of the loop spacing z and the permeability μ of the metal plate. The experimental work described later shows only a slight variation of S with z. (See Figs. 15–18.) Thus, within the limits $0.15 < r_1/a < 0.9$ for ten percent accuracy, the omission of z in (7) seems reasonable. Figures 2–4 are graphs of (7) for the two thicknesses of aluminum, copper, and steel used in the experimental work to follow. The experimental curves are those corresponding to loop spacing $z = 3$ cm. (See Fig. 14.)

Attention now turns to a solution of the problem by solving the vector wave equation for the vector potential:[2]

$$\nabla^2 \vec{A} + k_0^2 \vec{A} = 0. \quad (8)$$

The geometry of the problem suggests that the work might be done in cylindrical coordinates of unit vectors

[2] This problem has also been considered by R. K. R. Jonnada of the Moore School of Electrical Engineering, University of Pennsylvania, Philadelphia, Pa.

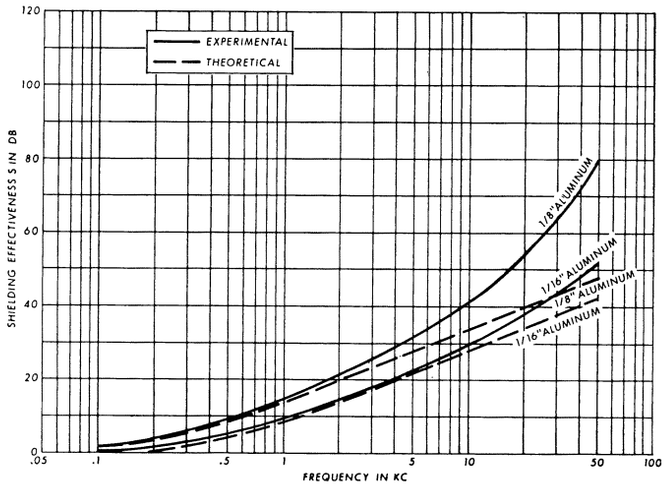


Fig. 2. Comparison of shielding effectiveness S for aluminum by two techniques.

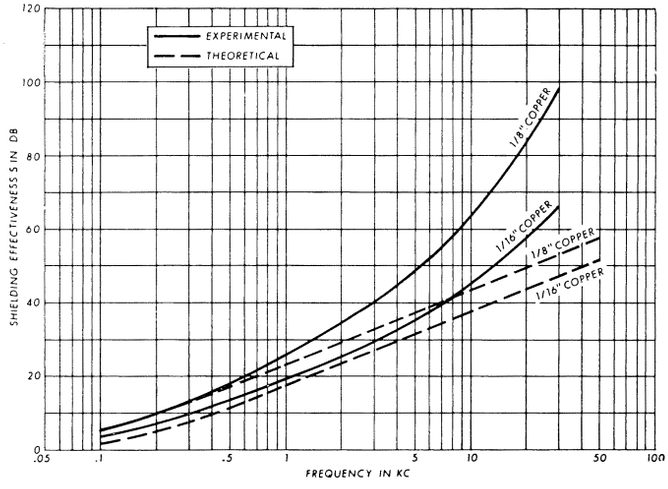


Fig. 3. Comparison of shielding effectiveness S for copper by two techniques.

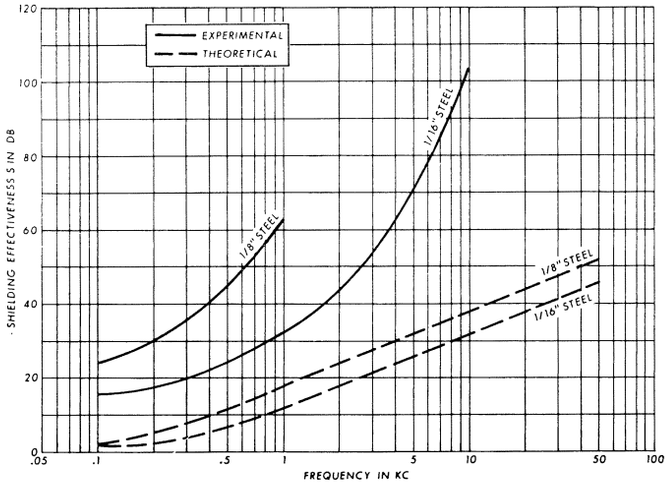


Fig. 4. Comparison of shielding effectiveness S for steel by two techniques.

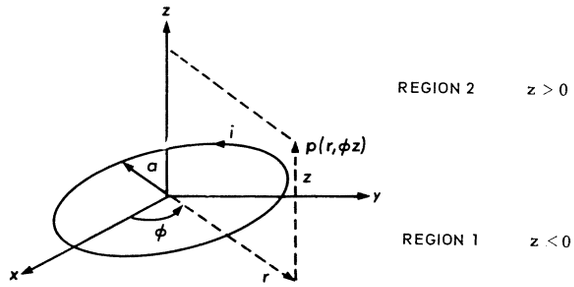


Fig. 5. Loop 1 (field source) in free space.

$\hat{\imath}_r$, $\hat{\imath}_\phi$, and $\hat{\imath}_z$, as shown in Fig. 5.

$$\vec{A} = A_\phi \hat{\imath}_\phi \tag{9}$$

$$\nabla \cdot \vec{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}.$$

But

$$A_r = 0 = A_z; \quad \frac{\partial}{\partial \phi} = 0 \text{ (cylindrical symmetry)} \tag{10}$$

therefore

$$\nabla \cdot \vec{A} = 0. \tag{11}$$

Similarly

$$\nabla \times \vec{A} = -\hat{\imath}_r \frac{\partial A_\phi}{\partial z} + \hat{\imath}_z \left\{\frac{\partial A_\phi}{\partial r} + \frac{1}{r} A_\phi\right\}. \tag{12}$$

Performing the curl again, we have

$$\nabla + \nabla \vec{A} = -\hat{\imath}_\phi \left\{\frac{\partial^2 A_\phi}{\partial z^2} + \frac{\partial^2 A_\phi}{\partial r^2} + \frac{1}{r}\frac{\partial A_\phi}{\partial r} - \frac{1}{r^2} A_\phi\right\}. \tag{13}$$

Using the identity

$$\nabla^2 \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla \times \nabla \times \vec{A}$$

the wave equation becomes

$$\frac{\partial^2 A_\phi}{\partial z^2} + \frac{\partial^2 A_\phi}{\partial r^2} + \frac{1}{r}\frac{\partial A_\phi}{\partial r} - \frac{1}{r^2} A_\phi + k_0^2 A_\phi = 0.$$

To apply the technique of separation of variables, let

$$A_\phi = R(r)Z(z). \tag{14}$$

With this substitution we obtain from the previous equation

$$\frac{\partial^2 Z}{\partial z^2} + (k_0^2 - \lambda^2)Z = 0 \tag{15}$$

$$\frac{\partial^2 R}{\partial r^2} + \frac{1}{r}\frac{\partial R}{\partial r} + \left(\lambda^2 - \frac{1}{r^2}\right) R = 0. \tag{16}$$

The solution of (15) is

$$Z(z) = Ae^{\tau_0 z} + Be^{-\tau_0 z} \tag{17}$$

where

$$\tau_0 = \sqrt{\lambda^2 - k_0^2}$$

and

$$k_0^2 = \omega^2 \mu_0 \epsilon_0.$$

The solution of (16) is

$$R(r) = C_1 J_1(\lambda r) + C_2 Y_1(\lambda r). \tag{18}$$

To satisfy the boundary conditions at $r = 0$ in (18) and at $z = \pm\infty$ in (17), let $C_2 = 0$ and write

$$A_\phi = C_1 J_1(\lambda r) \begin{cases} Ae^{\tau_0 z} & z < 0 \\ Be^{-\tau_0 z} & z > 0 \end{cases}$$

(taking the root of $\lambda^2 - k_0^2$ which has its real part greater than zero).

If the solution is to extend to infinity radially, there are no cylindrical boundaries to impose restrictions on the radial function $R(r)$, and therefore, λ has a continuous, rather than a discrete, spectrum. Thus, the solution involves an integral over all λ:[7]

$$A_\phi^{(1)} = \int_0^\infty B_1(\lambda) J_1(\lambda r) e^{\sqrt{\lambda^2 - k_0^2} z}\, d\lambda, \qquad z < 0 \tag{19}$$

$$A_\phi^{(2)} = \int_0^\infty B_2(\lambda) J_1(\lambda r) e^{-\sqrt{\lambda^2 - k_0^2} z}\, d\lambda, \qquad z > 0. \tag{20}$$

At $z = 0$, $A_\phi^{(1)} = A_\phi^{(2)}$; so $B_1(\lambda) = B_2(\lambda) = B(\lambda)$. Also,

$$\hat{n} \times (H_2 - H_1) = \text{true surface current} \tag{21}$$

where $\hat{n} = \pm \hat{i}_z$ and H_2 and H_1 are radial components of magnetic field intensity. Therefore, from (12) and the vector relation $\vec{B} = \nabla \times \vec{A}$,

$$\left.\frac{\partial A_\phi^{(1)}}{\partial z}\right|_{z=0-\epsilon} - \left.\frac{\partial A_\phi^{(2)}}{\partial z}\right|_{z=0+\epsilon} = \mu_0 i_d \delta(r - a). \tag{22}$$

Performing the indicated operations on (19) and (20) and letting ϵ approach zero, we obtain

$$2\int_0^\infty B(\lambda') J_1(\lambda' r) \sqrt{\lambda'^2 - k_0^2}\, d\lambda' = \mu_0 i_d \delta(r - a) \tag{23}$$

where λ' replaces λ as a dummy variable. We now multiply through by $J_1(\lambda r) r\, dr$ and integrate

$$2\int_0^\infty J_1(\lambda r) r\, dr \int_0^\infty B(\lambda') \sqrt{\lambda'^2 - k_0^2}\, J_1(\lambda' r) d\lambda' = \int_0^\infty \mu_0 i_d \delta(r - a) J_1(\lambda r) r\, dr. \tag{24}$$

We may make use of the Fourier–Bessel integral, which states that, with suitable restrictions on n and $f(x)$,[8]

$$f(x) = \int_0^\infty t J_n(tx) \left[\int_0^\infty \mu f(\mu) J_n(t\mu) d\mu\right] dt. \tag{25}$$

If we extract λ' from $\sqrt{\lambda'^2 - k_0^2}$ in (24) and let $\lambda' = \mu$, $r = t$, $x = \lambda$, (25) yields directly

$$B(\lambda) = \frac{\mu_0 a i}{2} \frac{\lambda}{\sqrt{\lambda^2 - k_0^2}} J_1(\lambda a). \tag{26}$$

Thus, we obtain

$$A_\phi^{(1)} = \frac{\mu_0 a i}{2} \int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{\tau_0 z}\, d\lambda, \; z < 0 \tag{27}$$

$$A_\phi^{(2)} = \frac{\mu_0 a i}{2} \int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a) J_1(\lambda r) e^{-\tau_0 z}\, d\lambda, \; z > 0. \tag{28}$$

These completely and uniquely describe the vector potential for all space surrounding the loop.

To check the results we use the well-known identity

$$\int_0^\infty e^{-\tau_0 z} J_0(\lambda a) \frac{\lambda}{\tau_0} d\lambda = \frac{e^{-jk_0 r}}{r} \tag{29}$$

where

$$r^2 = a^2 + z^2.$$

But

$$d/da\, [J_0(\lambda a)] = -\lambda J_1(\lambda a).$$

Therefore, taking the derivative of both sides of (29) with respect to a, we obtain

$$\left(jk_0 a + \frac{a}{\sqrt{a^2 + z^2}}\right) \frac{e^{-jk_0\sqrt{a^2+z^2}}}{a^2 + z^2} = \int_0^\infty \frac{\lambda^2}{\tau_0} e^{-\tau_0 z} J_1(\lambda a) d\lambda. \tag{30}$$

However, from (12),

$$B_z - (\nabla \times \vec{A})_z = \frac{\partial A_\phi}{\partial r} + \frac{1}{r} A_\phi = \frac{\mu_0 a i}{2} \int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a) J_0(\lambda r) e^{-\tau_0 z}\, d\lambda.$$

Therefore, at $r = 0$, we have $J_0(\lambda r) = 1$ for all finite λ and

$$B_z = \frac{\mu_0 a i}{2} \left(jk_0 a + \frac{a}{\sqrt{a^2 + z^2}}\right) \frac{e^{-jk_0\sqrt{a^2+z^2}}}{a^2 + z^2}. \tag{31}$$

For frequencies below 100 kHz and loop radii less than 5 cm, $k_0 a = \omega a \sqrt{\mu_0 \epsilon_0}$ is less than 10^{-4}. For $z < 9.9a$, $a/\sqrt{a^2 + z^2}$ is greater than 10^{-1}. Therefore, for the range of variables $f < 10^5$ Hz, $a < 5$ cm, and $z < 10a$, we see that the product $k_0\sqrt{a^2 + z^2}$ is less than 10^{-3}. Hence, we can write to a good approximation

$$B_z = \frac{\mu_0 a^2 i}{2(a^2 + z^2)^{3/2}} \tag{32}$$

which agrees with the result for magnetostatic flux density along the axis of a loop.[9]

Introducing the conducting plate as shown in Fig. 6, we have $A_\phi = A_\phi^{\text{impressed}} + A_\phi^{\text{induced}}$. The eddy current density within the plate has ϕ symmetry; therefore the induced vector potential will be the same formally as the impressed vector potential, modified at most by some function of λ:

$$A_\phi^{(1)} = \frac{\mu_0 ai}{2}\left\{\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)e^{-\tau_0|z|}\,d\lambda + \int_0^\infty C(\lambda)\frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)e^{\tau_0 z}\,d\lambda\right\} \quad (33)$$

$$A_\phi^{(2)} = \frac{\mu ai}{2}\left\{\int_0^\infty D(\lambda)\frac{\lambda}{\tau} J_1(\lambda a)J_1(\lambda r)e^{-\tau z}\,d\lambda + \int_0^\infty E(\lambda)\frac{\lambda}{\tau} J_1(\lambda a)J_1(\lambda r)e^{\tau z}\,d\lambda\right\} \quad (34)$$

$$A_\phi^{(3)} = \frac{\mu_0 ai}{2}\int_0^\infty F(\lambda)\frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)e^{-\tau_0 z}\,d\lambda \quad (35)$$

where

$$\tau = \sqrt{\lambda^2 - k^2}$$

and

$$k^2 = j\omega\mu\sigma.$$

At $z = b$,

$$A_\phi^{(1)} = A_\phi^{(2)}$$

$$\frac{1}{\mu_0}\frac{\partial A_\phi^{(1)}}{\partial z} = \frac{1}{\mu}\frac{\partial A_\phi^{(2)}}{\partial z}. \quad (36)$$

At $z = c$,

$$A_\phi^{(2)} = A_\phi^{(3)}$$

$$\frac{1}{\mu}\frac{\partial A_\phi^{(2)}}{\partial z} = \frac{1}{\mu_0}\frac{\partial A_\phi^{(3)}}{\partial z}. \quad (37)$$

Performing the indicated operations, letting $K = \tau/\tau_0$, $\mu_r = \mu/\mu_0$, and $t = c - b$, we obtain

$$\begin{array}{llll}
1) & Ke^{\tau_0 b}\,C(\lambda) & -\mu_r e^{-\tau b}\,D(\lambda) & -\mu_r e^{\tau b}\,E(\lambda) + 0 = -Ke^{-\tau_0 b} \\
2) & e^{\tau_0 b}\,C(\lambda) & +e^{-\tau b}\,D(\lambda) & -e^{\tau b}\,E(\lambda) + 0 = e^{-\tau_0 b} \\
3) & 0 & +\mu_r e^{-\tau c}\,D(\lambda) & +\mu_r e^{\tau c}\,E(\lambda) - KF(\lambda)e^{-\tau_0 c} = 0 \\
4) & 0 & -e^{-\tau c}\,D(\lambda) & +e^{\tau c}\,E(\lambda) + e^{-\tau_0 c}\,F(\lambda) = 0
\end{array} \quad (38)$$

$$\Delta = e^{-\tau_0(c-b)}\begin{vmatrix} K & -\mu_r e^{-\tau b} & -\mu_r e^{\tau b} & 0 \\ 1 & e^{-\tau b} & -e^{\tau b} & 0 \\ 0 & \mu_r e^{-\tau c} & \mu_r e^{\tau c} & -K \\ 0 & -e^{-\tau c} & e^{\tau c} & 1 \end{vmatrix} = Ce^{t(\tau-\tau_0)}$$

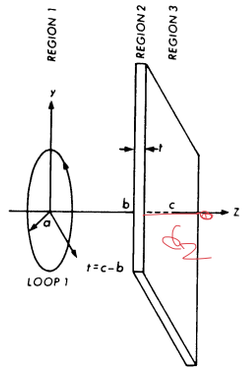


Fig. 6. Loop 1 and infinite shield plate of relative conductivity σ_r, relative permeability μ_r.

where

$$C = [(K + \mu_r)^2 - (K - \mu_r)^2 e^{-2t\tau}]^{-1}.$$

Solving similar determinants to find $C(\lambda)$, $D(\lambda)$, $E(\lambda)$, and $F(\lambda)$, we have

$$C(\lambda) = Ce^{-2b\tau_0}\{(K^2 - \mu_r^2)(e^{-2t\tau} - 1)\} \quad (39)$$

$$D(\lambda) = 2CKe^{b(\tau-\tau_0)}(K + \mu_r) \quad (40)$$

$$E(\lambda) = 2CKe^{-(t+c)\tau - b\tau_0}(K - \mu_r) \quad (41)$$

$$F(\lambda) = 4CK\mu_r e^{-t(\tau-\tau_0)}. \quad (42)$$

Substituting (39)–(42) into (33)–(35), we get

$$A^{\phi(1)} = \frac{\mu_0 ai}{2}\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)[e^{-\tau_0|z|} + Ce^{\tau_0(z-2b)}(K^2 - \mu_r^2)(e^{-2t\tau} - 1)]d\lambda \quad (43)$$

$$A_\phi^{(2)} = \mu ai\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)Ce^{b(\tau-\tau_0)}[e^{-\tau z}(K + \mu_r) + e^{\tau(z-2c)}(K - \mu_r)]d\lambda \quad (44)$$

$$A_\phi^{(3)} = 2\mu ai\int_0^\infty \frac{C\lambda\tau}{\tau_0^2} J_1(\lambda a)J_1(\lambda r)e^{-\tau_0 z - t(\tau-\tau_0)}\,d\lambda \quad (45)$$

where C is given by (38). These results are in agreement with those of Smythe[10] if we let $\omega = 0$.

The conditions set forth in (10) still hold when the shield is introduced; therefore, by the use of (12)

$$\vec{B}^{(1)} = \nabla \times \vec{A}^{(1)} = \frac{\mu_0 ai}{2}\left\{\hat{\imath}_r \int_0^\infty \lambda J_1(\lambda a)J_1(\lambda r)[e^{-\tau_0|z|} - Ce^{\tau_0(z-2b)}(K^2 - \mu_r{}^2)(e^{-2t\tau} - 1)]d\lambda + \hat{\imath}_z \int_0^\infty \frac{\lambda^2}{\tau_0} J_0(\lambda r)J_1(\lambda a)[e^{-\tau_0|z|} + Ce^{\tau_0(z-2b)} \times (K^2 - \mu_r{}^2)(e^{-2t\tau} - 1)]d\lambda\right\} \quad (46)$$

$$\vec{E}^{(1)} = -j\omega\vec{A}^{(1)} + \frac{\nabla(\nabla\cdot\vec{A}^{(1)})}{j\omega\mu\epsilon} = -j\omega\vec{A}^{(1)} \text{ [from (11)]} \quad (47)$$

$$\vec{E}^{(1)} = -\hat{\imath}_\phi \frac{j\omega\mu_0 ai}{2}\int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)[e^{-\tau_0|z|} + Ce^{\tau_0(z-2b)}(K^2 - \mu_r{}^2)(e^{-2t\tau} - 1)]d\lambda \quad (48)$$

$$\vec{B}^{(2)} = \mu ai\left\{\hat{\imath}_r \int_0^\infty \frac{\tau\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)Ce^{b(\tau-\tau_0)}[e^{-\tau z} \times (K + \mu_r) - e^{\tau(z-2c)}(K - \mu_r)]d\lambda + \hat{\imath}_z \int_0^\infty \frac{\lambda^2}{\tau_0} \times J_1(\lambda a)J_0(\lambda r)Ce^{b(\tau-\tau_0)}[e^{-\tau z}(K + \mu_r) + e^{\tau(z-2c)}(K - \mu_r)]d\lambda \quad (49)$$

$$\vec{E}^{(2)} = \overline{-j\omega\mu ai\hat{\imath}_\phi} \int_0^\infty \frac{\lambda}{\tau_0} J_1(\lambda a)J_1(\lambda r)Ce^{b(\tau-\tau_0)}[e^{-\tau z} \times (K + \mu_r) + e^{\tau(z-2c)}(K - \mu_r)]d\lambda \quad (50)$$

$$\vec{B}^{(3)} = 2\mu ai\left\{\hat{\imath}_r \int_0^\infty \frac{C\lambda\tau}{\tau_0} J_1(\lambda a)J_1(\lambda r)e^{-\tau_0 z - t(\tau-\tau_0)}\,d\lambda + \hat{\imath}_z \int_0^\infty \frac{C\lambda^2\tau}{\tau_0{}^2} J_1(\lambda a)J_0(\lambda r)e^{-\tau_0 z - t(\tau-\tau_0)}\,d\lambda\right\} \quad (51)$$

$$\vec{E}^{(3)} = -2j\omega\mu ai\hat{\imath}_\phi \int_0^\infty \frac{C\lambda\tau}{\tau_0{}^2} J_1(\lambda a)J_1(\lambda r)e^{-\tau_0 z - t(\tau-\tau_0)}\,d\lambda. \quad (52)$$

To obtain the free space flux density, we use (28) to obtain

$$\vec{B} = \nabla \times \vec{A} = \frac{\mu_0 ai}{2}\left\{\hat{\imath}_r \int_0^\infty \lambda J_1(\lambda a)J_1(\lambda r)e^{-\tau_0 z}\,d\lambda + \hat{\imath}_z \times \int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a)J_0(\lambda r)e^{-\tau_0 z}\,d\lambda\right\}.$$

Finally, we are in a position to write an expression for the shielding effectiveness of the infinite plate. For the loop orientation being considered, it is consistent with (7) to write

$$S = 20\log_{10}\left|\frac{B_z}{B_z{}^{(3)}}\right| \quad (53)$$

where B_z is the z-directed flux density at the same point on the z axis, but with the shield removed. Writing it out, we have

$$S = 20\log_{10}\frac{1}{4\mu_r}\left|\frac{\int_0^\infty \frac{\lambda^2}{\tau_0} J_1(\lambda a)e^{-\tau_0 z}\,d\lambda}{\int_0^\infty \frac{C\lambda^2\tau}{\tau_0{}^2} J_1(\lambda a)e^{-\tau_0 z - t(\tau-\tau_0)}\,d\lambda}\right|. \quad (54)$$

We could use (32) to simplify (54).

$$S = -20\log_{10}\frac{4\mu_r}{a}(a^2 + z^2)^{3/2}\left|\int_0^\infty \frac{C\lambda^2\tau}{\tau_0{}^2} \times J_1(\lambda a)e^{-\tau_0 z - t(\tau-\tau_0)}\,d\lambda\right|. \quad (55)$$

Equation (54) has no known closed form solution. Evaluation must be done by numerical integration on a computer. An attempt has been made to evaluate (54) with an IBM 704 computer. Although the numerator has converged to agree with (32), the numerical evaluation of the denominator requires great skill in programming to keep the calculations within the upper and lower machine limits. No results for (54) have been obtained to date.

We now consider the third analytical method of analyzing this problem. Schelkunoff considers a transmission line of length ℓ inserted between another line of propagation constant Γ_1 and characteristic impedance K_1, and a load impedance Z. He shows[3] that the transmission coefficient across the inserted uniform line of propagation constant Γ_2 and characteristic impedance K_2 is

$$T' = p'(1 - q'e^{-2r_2 l})^{-1} e^{-r_2 l} \quad (56)$$

where

$$p' = \frac{4K_1K_2}{(K_1 + K_2)(Z + K_2)}$$

and

$$q' = \frac{(K_2 - K_1)(K_2 - Z)}{(K_1 + K_2)(Z + K_2)}.$$

Since waves within a plane sheet of metal have field components almost completely in the plane of the shield[11] it is logical to make an analogy with a practical transmission line that can be almost entirely TEM. If we denote the wave impedance on either side of the shield by $Z_{\omega 1}$ and $Z_{\omega 3}$, and if we let $Z_{\omega 1}$ equal $Z_{\omega 3}$ (thus neglecting wave divergence in traversing our physically thin shield) and define the wave impedance and propagation constant within the shield to be η and $\bar{\sigma}$, respectively, one can argue by analogy that the transmission coefficient across the inserted homogeneous shield of thickness t is

$$T = p(1 - qe^{-2\bar{\sigma}t})^{-1} e^{-\bar{\sigma}t} \quad (57)$$

[3] Schelkunoff,[4] pp. 223–225.

where

$$p = \frac{4k}{(k+1)^2}, \quad q = \frac{(k-1)^2}{(k+1)^2}$$

and k is the ratio η/Z_ω or Z_ω/η, Z_ω being the incident wave impedance normal to the shield. Then the insertion loss in decibels between two corresponding points just outside the plate would be for the transverse field components

$$\begin{aligned} S &= -20 \log_{10} |T| \\ &= 20 \log_{10} \left|\frac{1}{p}\right| - 20 \left(\frac{1}{2.30259} \ln \left|e^{-\bar{\sigma}t}\right|\right) + \\ &\qquad 20 \log_{10} \left|1 - qe^{-2\bar{\sigma}t}\right| \qquad (58) \\ &= R + A + B \text{ in decibels.} \end{aligned}$$

Thus, in general, we must calculate

$$R = 20 \log_{10} \frac{|k+1|^2}{4|k|} \text{ in dB} \qquad (59)$$

$$A = -8.686 \ln \left|e^{-(1+j)\alpha t}\right| = 8.686\, \alpha t \text{ in dB} \qquad (60)$$

$$B = 20 \log_{10} \left|1 - qe^{-2\bar{\sigma}t}\right| \text{ in dB} \qquad (61)$$

where

$$\bar{\sigma} = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \text{ and } \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}.$$

For any good shield in the frequency range of interest $\sigma/\omega\epsilon \gg 1$. Therefore, by choosing the root of j that lies in the first quadrant (since this choice is arbitrarily but consistently made in the literature), these become

$$\bar{\sigma} = (1+j)\sqrt{\pi f\mu\sigma} \equiv (1+j)\alpha, \quad \eta = (1+j)\sqrt{\pi f\mu/\sigma}.$$

Thus, R represents the reflection loss due to the initial reflection at both interfaces of the shield, B accounts for all reflections not contained in R, and A is the attenuation of the field components in penetrating the shield once. The most difficult component to calculate is B, but fortunately, as A gets large, B becomes negligible. Usually B is neglected for values of A in excess of 15 dB. The reader is referred to Schelkunoff for detailed discussion of this theory.

Some remarks should be made concerning the impedance of the wave which is incident onto the shield. An exact description of the fields in the immediate vicinity of the source (loop 1) would involve expressions containing complete elliptic integrals of the first and second kind. One would be hard put to apply meaningfully the resulting expression for the incident wave impedance Z_ω in (59) and (61). There is reason to believe that the fields about loop 1 do not differ markedly from those from a similarly oriented elemental magnetic dipole. Forming the ratio of E_θ to H_ϕ in the elemental dipole formulas[4] and considering the nearfield where $2\pi\rho/\lambda \ll 1$, ρ being the radial distance from the center of the dipole, one obtains for the radial wave impedance at ρ

$$Z_\omega = j\omega\mu_0\rho. \qquad (62)$$

We shall use this approximation for Z_ω of loop 1.

In our problem one would not, in general, expect (58) to predict the correct insertion loss created by placing the shield arbitrarily between the two loops. Consider the case when A exceeds 15 dB, thereby making B negligible. The analyses leading to (7) and (54) indicate that the insertion loss S is not a function of the shield position between the two fixed loops. By the theory just outlined A would be easily calculable by knowing the intrinsic properties of the shield and the source frequency of oscillation. But by (62) it is not correct to assume that R would be independent of shield position. The incident wave impedance, and hence R, would vary as one proceeded radially away from the z-axis along the source-side interface of the shield. It would seem that this variation would generally become larger as the loop 1 to shield spacing b decreased. But at loop 2, the shielded point of observation, the integrated effect of these circles of constant R would have to be independent of the shield position if the theories are to be consistent. Therefore, we should be able to virtually displace the shield between the source point and the point of observation for calculation purposes. Imagine the shield to be displaced to coincide with loop 2. Thus, to evaluate (58), we would replace ρ by the loop-to-loop spacing z in (62). Since the longitudinal field components are reduced by the same amount as the transverse components when the shield is present, one might expect (58) to predict the insertion loss created by the introduction of the shield between the two loops. The accuracy would be expected to increase as the loop-to-loop spacing increased.

Figures 7–12, which are graphs of shielding effectiveness S vs. frequency, were computer derived from (58) [using (61)]. Three materials were chosen for test: one having relative permeability and relative conductivity equal to unity (annealed copper); one having μ_r but not σ_r equal to unity (5052 aluminum alloy $\sigma_r = 0.35$); and one having neither μ_r nor σ_r equal to unity (steel, $\sigma_r = 0.17$, μ_r calculated to be 112). Each shield was evaluated for two thicknesses, $t = 1/16$ inch and $t = 1/8$ inch (corresponding to the six plates to be considered later by experimental laboratory techniques).

Comparing these data with the corresponding data obtained experimentally (Figs. 15–18), it appears that agreement is not as predicted by our proposed virtual displacement technique; differences generally tend to increase with increase in loop spacing. If one considers the plate to be virtually displaced such that we have (see Fig. 1)

$$\frac{r_1}{r_1 + r_2} = \frac{b}{z} = \frac{1}{3} \qquad (63)$$

careful comparison of corresponding curves suggests that (58) is useful in predicting the total insertion loss of the aluminum and copper shields within 3 dB. It should be stressed that (63) is based on laboratory obtained shielding effectiveness data which are not nearly as extensive

[4] Schelkunoff[4] p. 163.

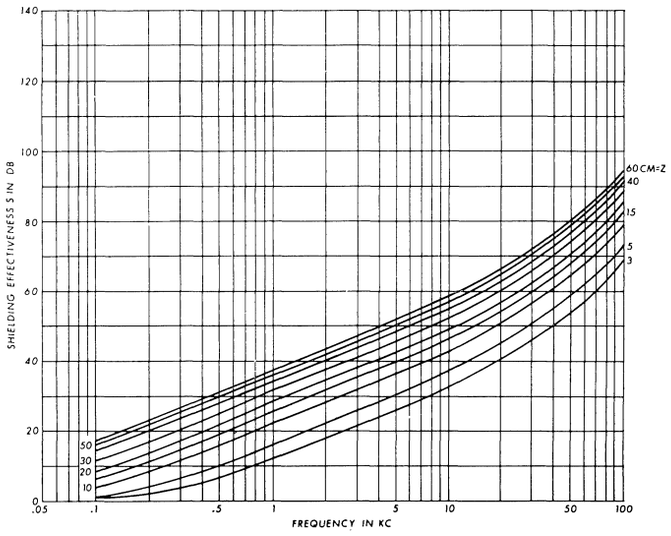


Fig. 7. Theoretical shielding effectiveness S vs. frequency by Schelkunoff's $S = R + A + B$ for 1/16 inch aluminum.

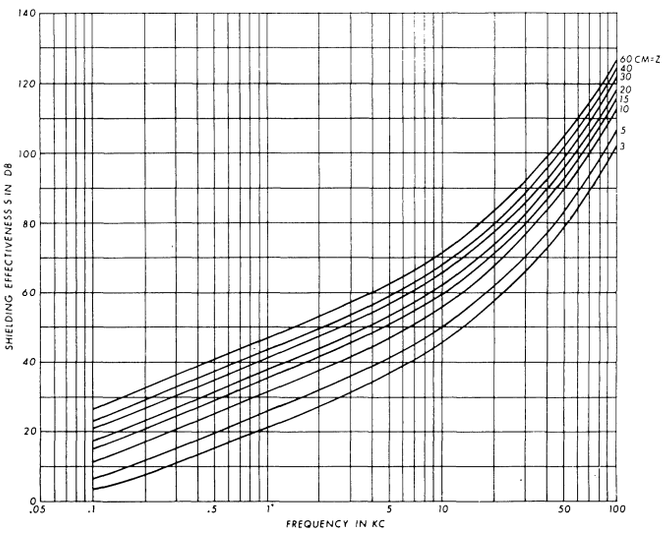


Fig. 8. Theoretical shielding effectiveness S vs. frequency by Schelkunoff's $S = R + A + B$ for 1/8 inch aluminum.

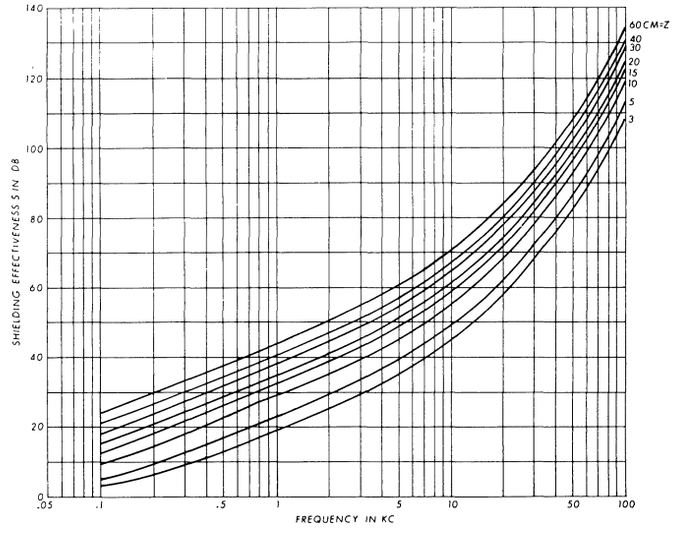


Fig. 9. Theoretical shielding effectiveness S vs. frequency by Schelkunoff's $S = R + A + B$ for 1/16 inch copper.

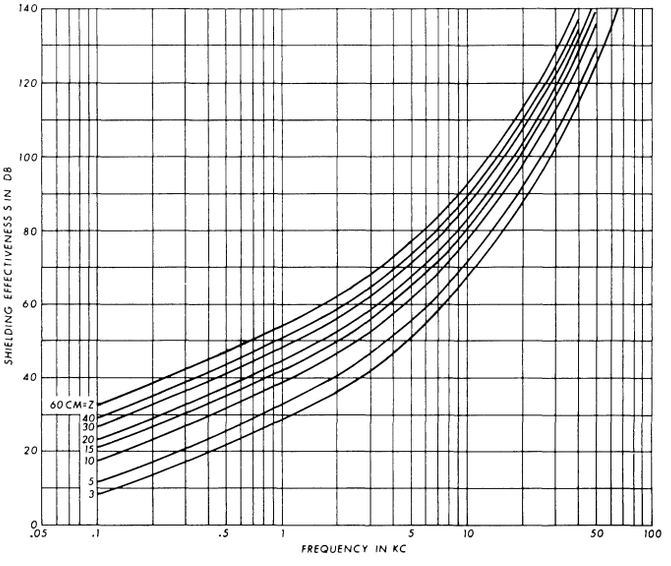


Fig. 10. Theoretical shielding effectiveness S vs. frequency by Schelkunoff's $S = R + A + B$ for 1/8 inch copper.

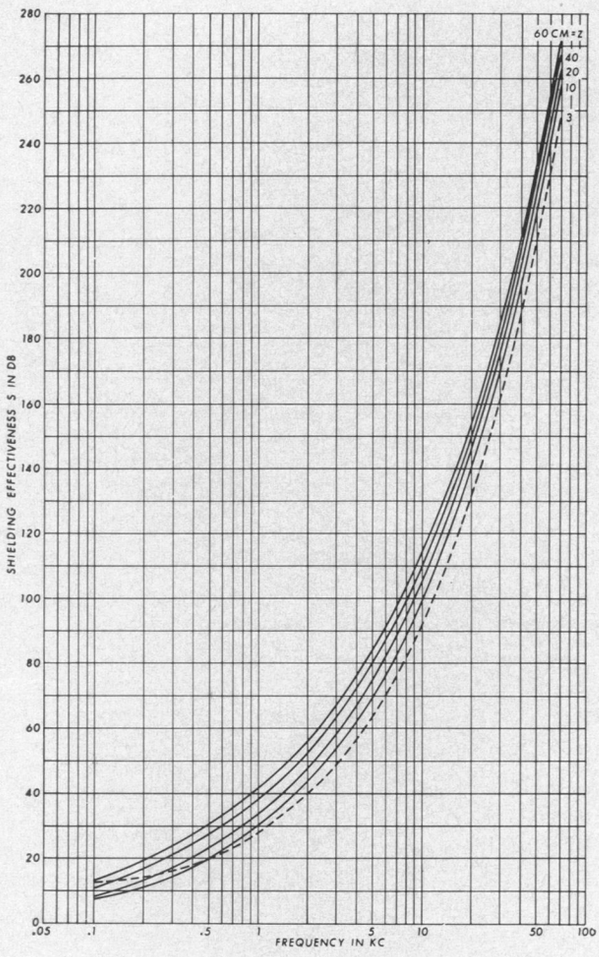


Fig. 11. Theoretical shielding effectiveness S vs. frequency by Schelkunoff's $S = R + A + B$ for 1/16 inch steel.

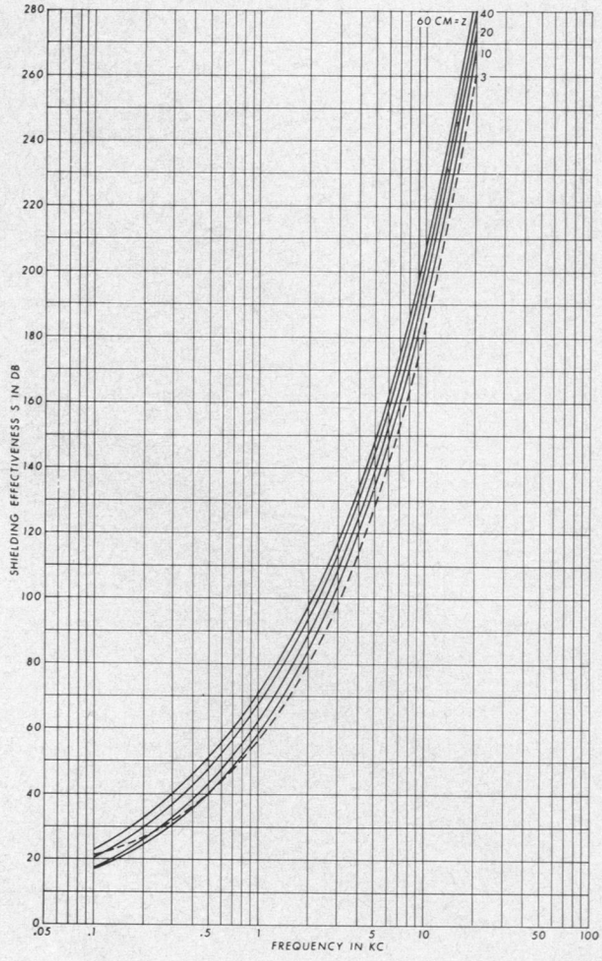


Fig. 12. Theoretical shielding effectiveness S vs. frequency by Schelkunoff's $S = R + A + B$ for 1/8 inch steel.

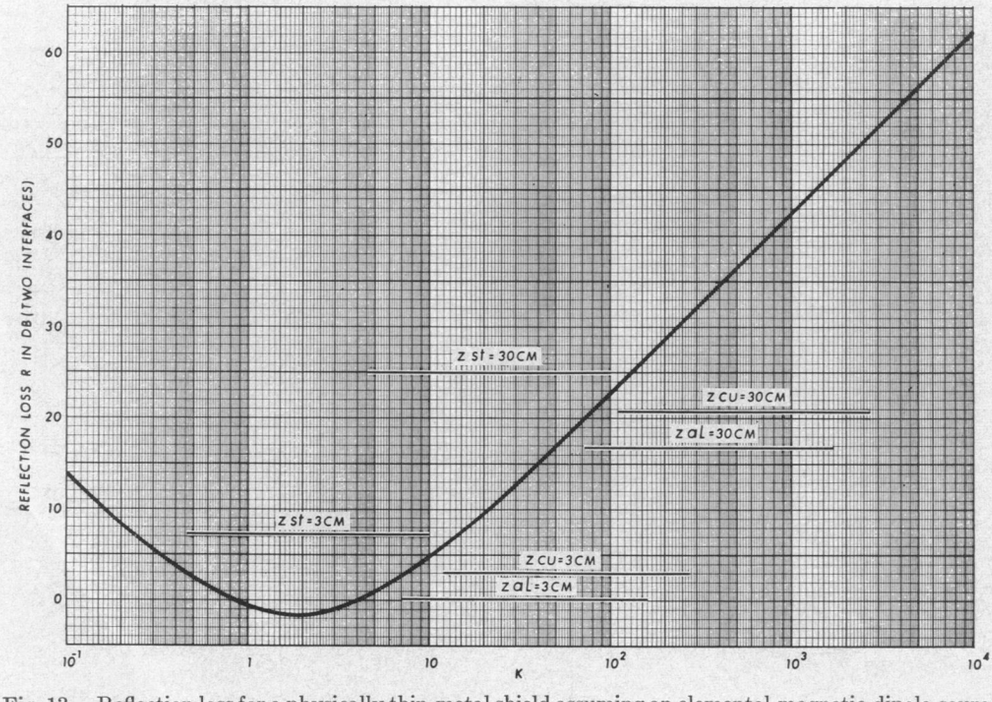


Fig. 13. Reflection loss for a physically thin-metal shield assuming an elemental magnetic dipole source. $R = 20 \log_{10} [0.462K^{-1} + 0.136K + 0.354]$ where $K = (\sqrt{f\mu_r/\sigma_r})z$ and z = radial distance of magnetic dipole from shield in inches.

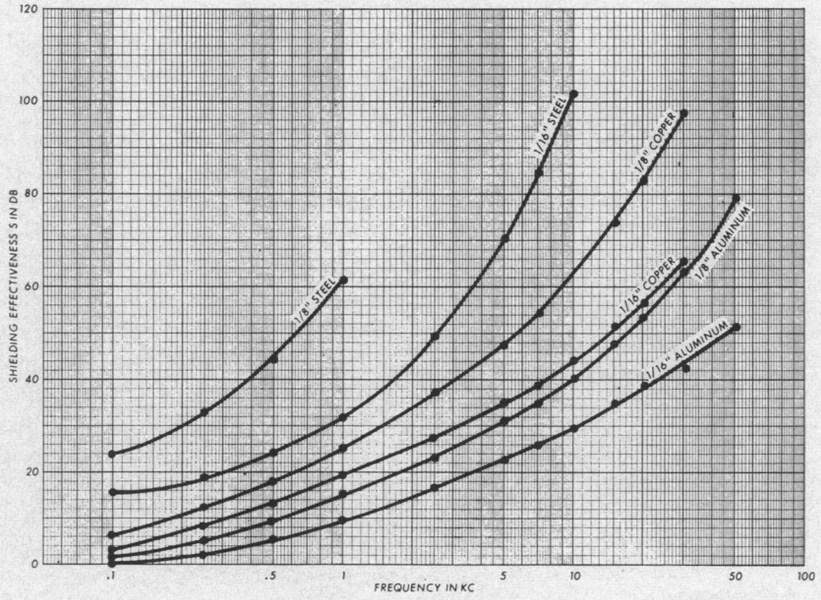


Fig. 14. Experimental insertion loss S vs. frequency for the six shields considered. (All curves for loop 1 radius = 3.5 cm, loop spacing z = 3 cm.)

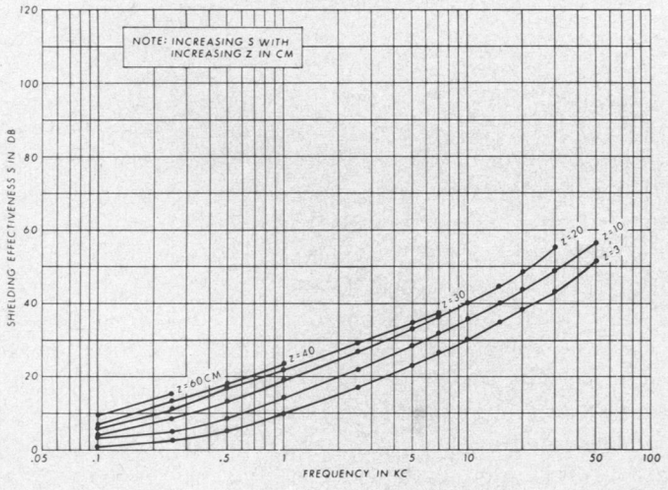


Fig. 15. Experimental shielding effectiveness S vs. frequency for 1/16 inch aluminum.

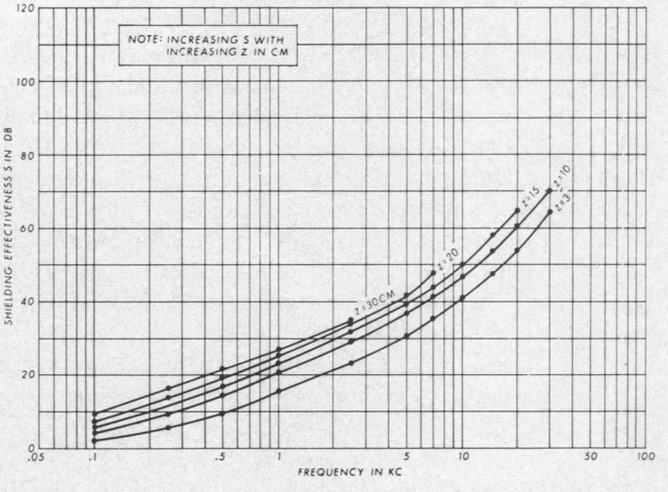


Fig. 16. Experimental shielding effectiveness S vs. frequency for 1/8 inch aluminum.

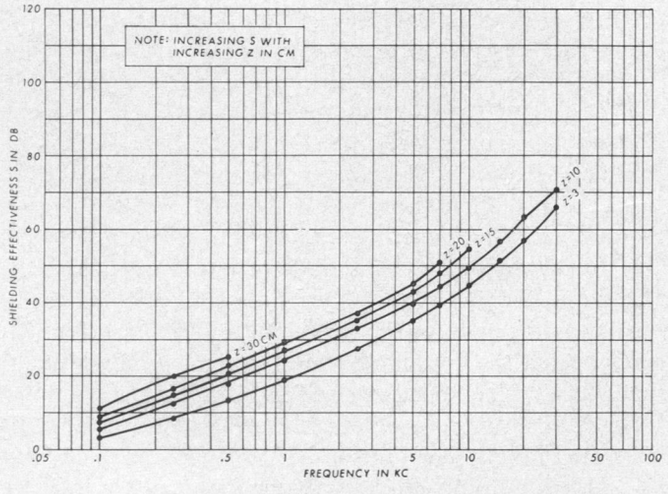


Fig. 17. Experimental shielding effectiveness S vs. frequency for 1/16 inch copper.

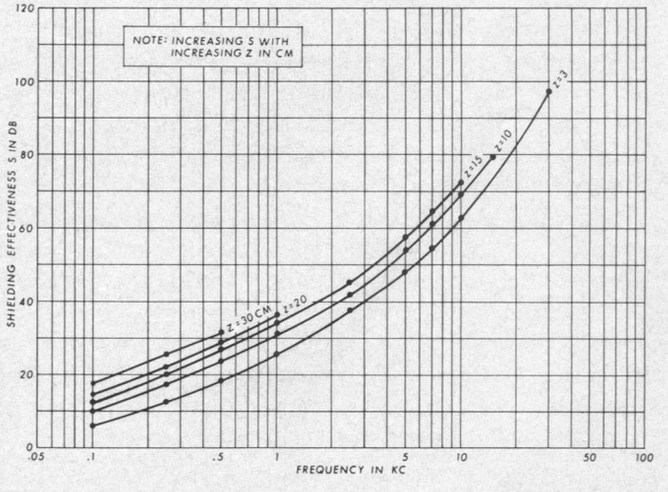


Fig. 18. Experimental shielding effectiveness S vs. frequency for 1/8 inch copper.

as the computer results. Thus, one should be hesitant to use (63) beyond the range of the experimental data.

Figure 13 is included as a convenient graphical aid in evaluating (59) when a computer is not available. In Figs. 7–12 and 14–18 the parameter z represents the loop 1 to point of observation axial distance.

Experimental Data

Last in our discussion is the experimental solution of the problem. A test arrangement that allowed the two coaxial loops to be kept at a constant separation z while a 4-foot by 4-foot shield was moved between them from one extreme to the other was constructed. Loops of 3.5 cm radius and 7 turns were used for loops 1 and 2 of Fig. 1. The independence of S and the loop 1 to shielded distance b predicted in (7) and (54) has been demonstrated experimentally for different shielding materials and thicknesses. This provided the experimenter with a means of detecting data contaminated by end effects. When a variation of S with shield position between the two loops was observed, the data were not taken as a true representation of the shielding effectiveness of the shield being tested. It was consistently observed that end effects seem to become more and more pronounced with increasing frequency, shielding effectiveness of the shield, and loop spacing z, i.e., for the more effective shields, end effects occurred at much lower frequencies for the same loop spacing. Figures 14–18 graphically illustrate the shielding effect vs. frequency curves for the six plates tested. All data not contaminated by end effects are included.[5] Thus, less meaningful shielding effectiveness data are obtainable with the more effective shields, as loop spacing or frequency is increased. An explanation of this phenomenon follows:

1) The leakage fields are assumed to be independent of the frequency. They should be relatively constant for relatively small variation in loop spacing z since they are mainly a function of the leakage path length around the shield. But the leakage voltage V_L induced in loop 2 will be a linear increasing function of frequency if the field and loop positions are held constant.

2) From Fig. 14 the direct coupling voltage V_D to loop 2 goes down (between 1 and 50 kHz for the materials tested) by a complex function of frequency. Therefore, one can see that the ratio V_L/V_D is an increasing function of frequency, shielding effectiveness, and loop spacing z. It is this ratio which should be kept small when shielding effect measurements are being conducted.

Figures 15–18 demonstrate the small but persistent dependence of S upon loop spacing z. It is interesting to note that with the aluminum and copper shields the shielding effectiveness always increased as loop spacing z was increased. However, the shielding effectiveness either increased or decreased with increasing loop spacing for the steel plates, depending on the loop spacing and frequency. Arguing, as we did previously, that the shielding effectiveness S is to be calculated by virtually displacing the shield to coincide with loop 2, one can justify these results on the basis of (59), i.e., as loop spacing increases, Z_ω increases, k changes, and the initial reflection loss changes. Figure 13 is a plot of (59) where an elemental loop dipole of radial impedance $j\omega\mu_0 z$ is the field source. The horizontal lines represent the range of K for each of the three materials tested. The apparent anomaly in the data is then easily justified. The reflection loss R for the steel plates with small values of loop spacing z goes through a minimum as frequency is varied from 100 Hz to 50 kHz. See also Figs. 11 and 12.

Figure 19 represents shielding data obtained with the two loop axes parallel to the plane of the shield interfaces. From the data obtained by using the 1/16-inch aluminum shield, the shielding does not differ greatly from that observed with the original loop orientation, especially for the larger loop separations. This strongly supports the assumption that the wave impedance can be calculated on the basis of an elemental magnetic dipole at radial distances less than would seem justifiable.

Conclusion

Figures 2 and 3 show that the experimental and theoretical results obtained by the first analytical method are within 3 dB for values of shielding effectiveness below 25 dB for the aluminum and copper sheets. Agreement seems to get better as the frequency is reduced. Inspection of Fig. 4 implies that (7) is not accurate for shielding effectiveness determination of sheets with relative permeability much different from unity. Thus, the neglected quantities (magnetic permeability and axial variation of eddy currents within the sheet) tend to limit the usefulness of our first method for solving this problem.

The second method, although providing a theoretically exact formulation, is the least practical since it requires a computer solution which has not yet been carried out. A table of shielding effectiveness for various materials, thicknesses, loop dimensions, spacing, and frequencies resulting from such a solution would be extremely useful. Using this method we were able to predict the independence of S and loop 1 to shield spacing b.

The third analytical technique is adapted to our problem from Schelkunoff's well-known transmission theory of shielding. By direct comparison of the theoretical data (which are computer-derived from the results of this theory) with the experimentally obtained data, we are able to determine the usefulness of our third analytical technique. Figures 20–27 show some comparisons of the corresponding data for each material tested. From Fig. 13, one sees that the initial reflection loss for steel is relatively small for the loop spacings and frequencies at which experimental data were obtainable. All experimental data for steel fell within the region between the $z = 3$ cm and $z = 60$ cm shielding effect vs. frequency curves of Figs. 11 and 12. The theoretical curves of Figs. 20–27 represent

[5] Experimentally obtained shielding data for steel are presented for loop spacing $z = 3$ cm (see Fig. 14). A very limited amount of data for larger loop spacings was obtained, but clear graphical presentation of all these data was not possible. Some additional shielding data are found in Figs. 26 and 27.

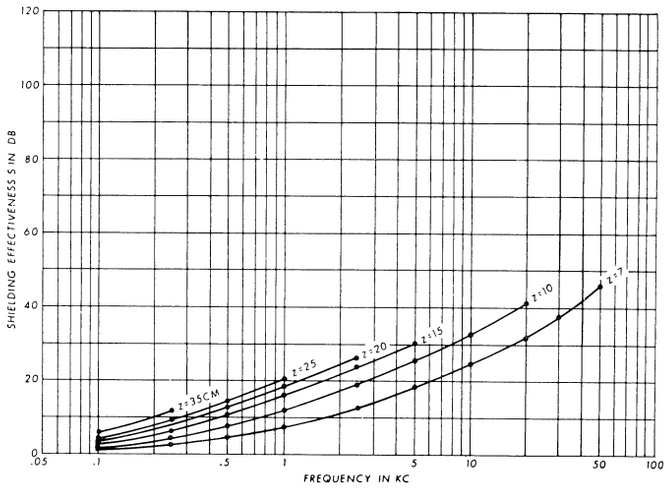


Fig. 19. Shielding effectiveness S vs. frequency for 1/16 inch aluminum. (Loop axes parallel to shield loop center-to-center distance z in cm.)

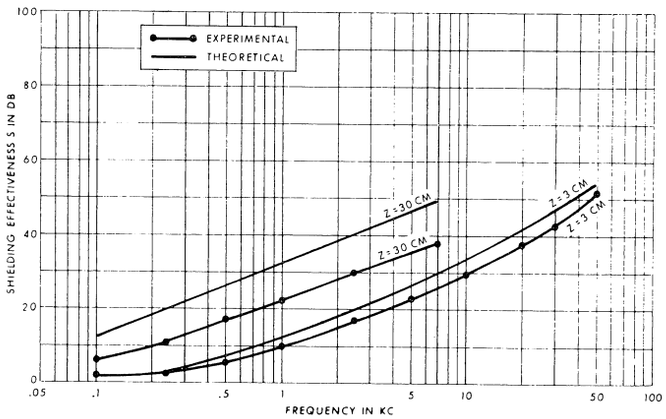


Fig. 20. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/16 inch aluminum.

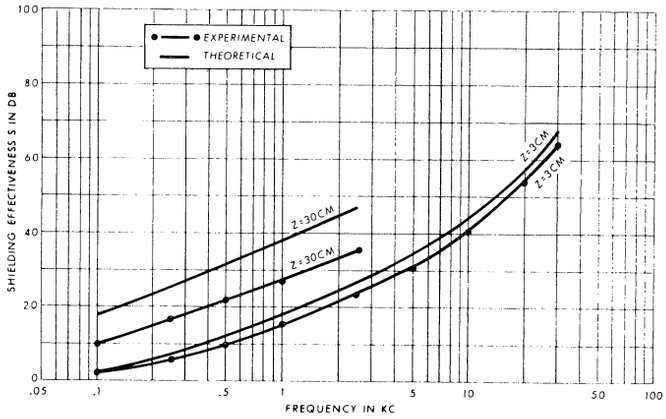


Fig. 21. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/8 inch aluminum.

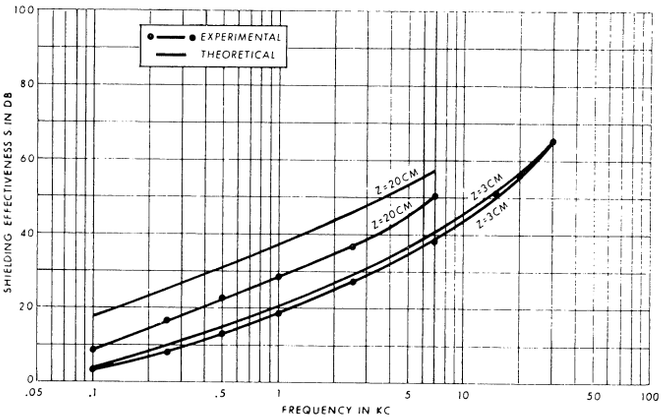


Fig. 22. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/16 inch copper.

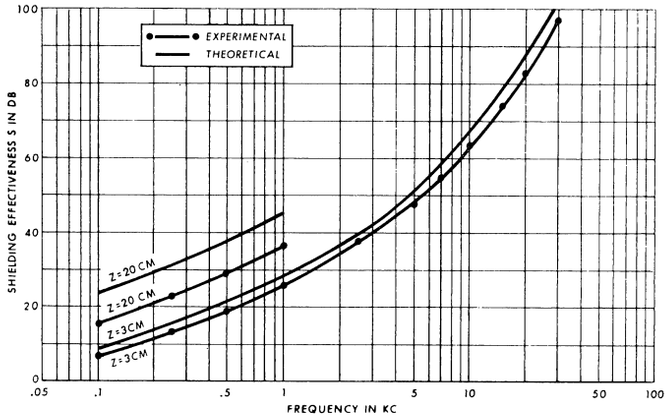


Fig. 23. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/8 inch copper.

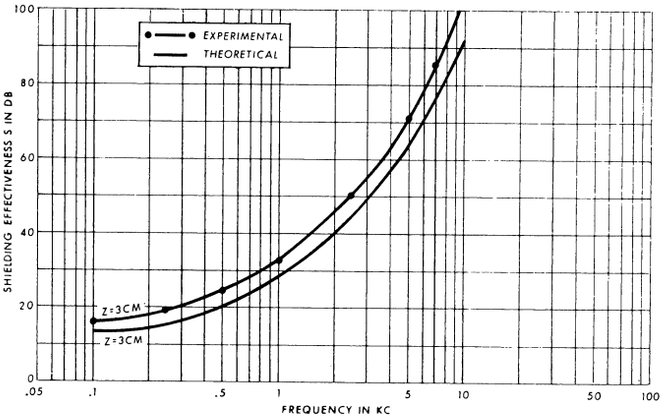


Fig. 24. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/16 inch steel, loop spacing $z = 3$ cm.

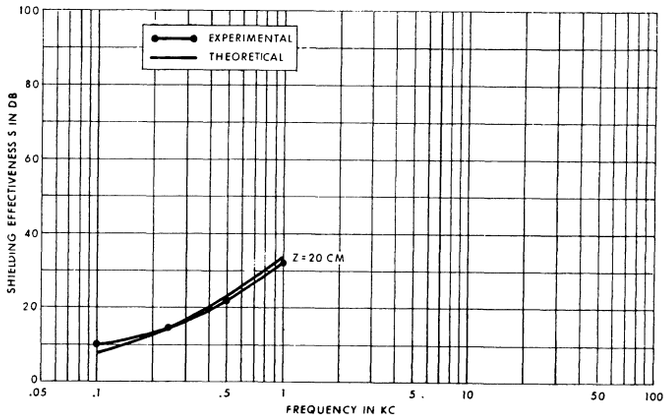


Fig. 25. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/16 inch steel, loop spacing $z = 20$ cm.

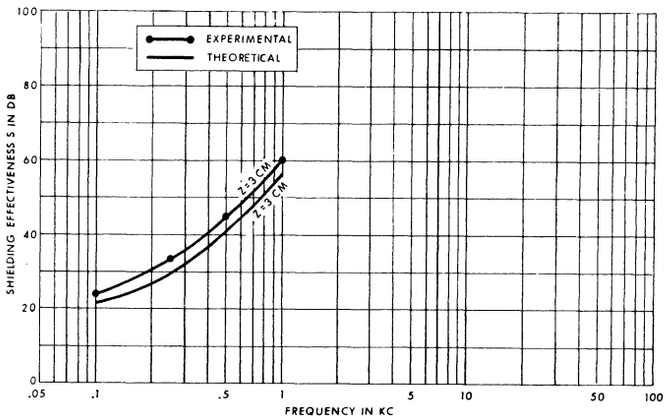


Fig. 26. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/8 inch steel, loop spacing $z = 3$ cm.

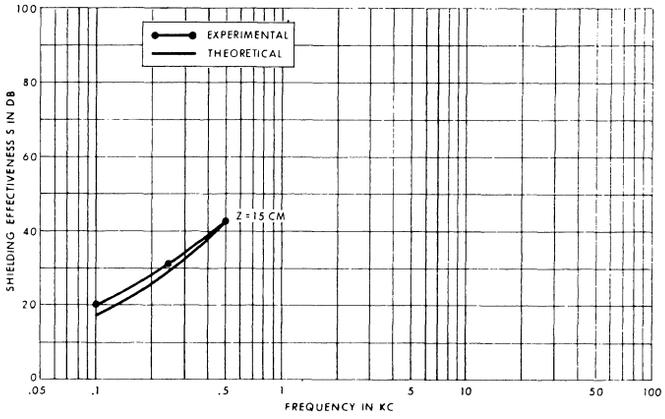


Fig. 27. Comparison of theoretical shielding effectiveness $S = R + A + B$ with experimentally obtained shielding effectiveness for 1/8 inch steel, loop spacing $z = 15$ cm.

shielding effectiveness $S = R + A + B$ calculated on the premise that the shield is virtually displaced to coincide with the shielded point of observation. Agreement between experimental and theoretical values of shielding effectiveness for aluminum and copper is attained within 3 dB provided the shield is placed (for calculation) a distance from the source equal to one-third of the loop spacing z.

Further evaluation of the transmission theory of shielding for the problem considered here is inhibited by the lack of a wider range of meaningful experimental data. More data could be obtained by increasing the plate size, or by employing other methods to reduce end effects.

This analytical technique is more powerful than our first analytical technique since it allows for a greater range of variables z and μ and shielding effectiveness S. Even after (54) is computer evaluated, thus providing us with a presumably exact solution for the problem, the shielding theory of Schelkunoff should be useful.

Acknowledgment

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